

Victor Nicolet

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About

Applied scientist working on problems ranging from static analysis, to anomaly detection and code generation. My research focuses on extending program synthesis techniques to solve interesting and difficult programming problems.

Experience

Applied Scientist, Amazon Web Services, Arlington, Virginia Aug. 2022–present
Working on static analysis, anomaly detection, and code generation.

- **Cedar** — open-source language for defining permissions as policies.
- **Argot** — lightweight collection of static analysis tools for Go, built on top of the standard Go analysis packages. Includes taint analysis, backtracing, reachability, and more.

Graduate Research Assistant, University of Toronto, Canada Sep. 2017–Aug. 2022
Research assistant to Prof. Azadeh Farzan. Participated in research projects, developing the theoretical framework as well as the supporting software artifacts.

- **Synduce** — synthesizes recursive functions given a functional specification, using selective bounding techniques and off-the-shelf solvers.
- **Parsynt** — synthesizes divide-and-conquer implementations of loops with equivalence proofs, outputting parallel C++ (Intel TBB) with verification in Dafny.

Research Intern, École Normale Supérieure, Team Parkas, Paris, France Mar.–Jul. 2015
Supervised by Albert Cohen. Studied a parallel execution environment based on Kahn Process Networks; developed a benchmark to analyze runtime performance.

Education

PhD, Computer Science, University of Toronto, Canada 2017–2022
Advisor: Azadeh Farzan. Research on formal methods and synthesis techniques applied to automatic parallelization.

Master of Science in Engineering, École Polytechnique, Palaiseau, France 2012–2016
Common core includes physics, informatics and mathematics. Specialized in Computer Science.

Parisian Master of Research in Computer Science (MPRI), Université Paris Diderot 2015–2016
Pursued jointly with second year of MSc at Telecom Paristech. Core training in the fundamentals of computer science and introduction to research.

Classes Préparatoires, Lycée aux Lazaristes, Lyon, France 2010–2012
Intensive undergraduate studies to prepare competitive exams. Specialized in Physics and Chemistry.

Publications

- [1] **EventADL: Open-Box Anomaly Detection and Localization Framework for Events in Cloud-Based Service Systems**
with L. Pham, J. Dodds, H. Guan, D. Kroening. *FSE 2026*.
- [2] **Hypergraph-Guided Regex Filter Synthesis for Event-Based Anomaly Detection**
with M. Ferreira, L. Pham, J. Dodds, D. Kroening, I. Lynce, R. Martins. *arXiv 2025*.
- [3] **Program Synthesis From Partial Traces**
with M. Ferreira, J. Dodds, D. Kroening. *PLDI 2025*.

- [4] **Recursion Synthesis with Unrealizability Witnesses**
with A. Farzan, D. Lette. *PLDI 2022*.
- [5] **Counterexample-Guided Partial Bounding for Recursive Function Synthesis**
with A. Farzan. *CAV 2021*.
- [6] **Phased Synthesis of Divide-And-Conquer Programs**
with A. Farzan. *PLDI 2021*.
- [7] **Modular Divide-And-Conquer Parallelization of Nested Loops**
with A. Farzan. *PLDI 2019*.
- [8] **Synthesis of Divide and Conquer Parallelism for Loops**
with A. Farzan. *PLDI 2017*.

Service

- POPL 2027 — Program Committee
- PLDI 2026 — Review Committee
- OOPSLA 2022, 2023 — External Review / Artifact Evaluation Committee
- VMCAI 2022 — Artifact Evaluation Committee
- PLDI 2018 — Artifact Evaluation Committee
- Dagstuhl Seminar 18111 on Loop Optimization